

Chapter 1

Short Manual Audioanalyser R&S UPV

You can get detailed information in D:\UPV\manuals on the UPV.

The UPV has a generator and analyzer function. Usually, to record the amplitude response, the generator is used in autosweep mode, i.e. the frequency is automatically increased from the start to the stop value. This signal is given to the input of the laboratory board.
If the output of the board is connected to the analyzer, the amplitude response is displayed.

Start the UPV and load a setfile:
For the labs there are preset setfiles on the device under D:\UPV\User\setups depending on the hardware to be measured, one of which must first be loaded:

- Switch on the UPV. Windows and subsequently the firmware start automatically.
- Press the gray 'Menu' button so that the menu appears at the top of the screen.
- Select `File` → `Load Setup` from the menu.
- Never save modified setfiles!

Screens:
The toggle switch 'Screens' is used to switch between the individual screens. Screen 1 contains the configurations for generator and analyzer in the given setfiles, screen 2 the output of the amplitude response.

Connect the UPV:
The generator has to be connected to the input, the analyser to the hardware output.
Terminators are not needed!

Measure a frequency response:
With the loaded setfile no further settings are necessary. After pressing the buttons 'Stop' and 'Start' on the left side of the CONTROL area of the keypad the amplitude response is output on Screen 2 (sweep function). The output window can be adjusted with right mouse button → *Config*.

Note for measuring the **phase response**: The signal to be measured is fed into Analyser2, Analyser1 must receive the reference signal (e. b. from the 2nd generator).

FFT of a signal:

If the spectrum of a signal is to be measured, e. g. at the output of the board, then on Screen 1 must be set for the analyzer (right column): *Analyser Function→Function: FFT* and the following box must be checked: Post FFT→Show. The FFT is immediately displayed in a separate window on this screen.

Save screenshot:

Usually you can save a screenshot with the button 'H COPY' (area CONTROL in the keyboard). This is then called D:\<Setfilename>.bmp and is overwritten without warning! So please rename before each new screenshot. The data can be transferred via USB stick (sockets on the right side in the front of the device).
If this way does not work (Setfile is wrong), you can create a screenshot with <Alt>+<Print/Druck>, which can be inserted and saved with <Ctrl>+<v> in MS Paint.

Switch off:

The firmware of the audio analyzer runs under Windows. Before switching off, Windows must be shut down as usual. After that, the power off switch must be pressed.

Last remark:

Please make sure that above the button *'OUTPUT'* (CONTROL area in the keypad) the LED next to *'OFF'* is not lit, otherwise there will be no output signal.

Chapter 2

Operating the UPV66

The UPV66 has no control panel and no own display. However, the hardware is the same, all functions can also be covered by this device.
The following tables will help you with the operation:

Die Funktionen der Maus entsprechen folgenden Frontplattenfunktionen:

Maus	**Frontplatte**	**Funktion**
Klick mit linker Maustaste	Taste ENTER, Drehradklick	• Aktivieren von Auswahlfeldern • Auswahl von Parametern • Aktivieren/deaktivieren von Tick-Boxen • Umschalten von Radio-Buttons, • Verlassen des Edit-Modes
Mauszeiger an den linken Rand bewegen	Taste MENU .	• Einblenden der Menü-Leiste
Mauszeiger an den rechten Rand bewegen	---	• Einblenden der Toolbar
Betätigen des Maus-Rädchens	Drehen am Drehknopf	• Navigieren in Panels, Parameterlisten, etc. • Verändern von Zahlenwerten
Drücken des Maus-Rädchens	Drehknopfklick	• Auswahl von Parametern • Verlassen des Edit-Modes

Figure 2.1: Mouse functions

Die Tasten auf der Frontplatte des Audio Analyzers entsprechen folgenden Tasten bzw. Tastenkombinationen auf der externen PC-Tastatur:

Frontplatte	PC-Tastatur		Frontplatte	PC-Tastatur
Softkeys F5 ... F12	F5 ... F12		0 ... 9; A ... Z	0 ... 9; A ... Z
START	CTRL + F5			SPACE
SINGLE	CTRL + F6		• ; * ... #	• ; * ... #
STOP CONT	CTRL + F7		+/-	+/-
Lautsprecher ON	CTRL + F8		A <-> a	SHIFT + A ... Z
Lautstärke	CTRL + F9		MENU	CTRL + M
H COPY	CTRL + P		UNDO	BACKSPACE
SHUT-DOWN	CTRL + F10		ESC	ESC
LOCAL	CTRL + F11		Enter, Drehradklick	ENTER
OUTPUT OFF	CTRL + F12		Cursor-Tasten	Cursor-Tasten
SCREEN <	CTRL + PAGE UP		WINBAR	ALT + W

Figure 2.2: Keyboard shortcuts